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ARTIFICIAL INTELLIGENCE AND COMPUTER SCIENCE

Design of a Data Warehouse for ATP Tennis Matches

Course: DATA WAREHOUSE AND VISUALIZATION

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1 Introduction

This project focuses on the design and implementation of a data warehouse for the analysis of professional tennis matches.

The starting point is the *ATP Tennis Daily Pull* dataset available on Kaggle. The version used in this project contains 67,512 matches played from 3 January 2000 to 12 April 2026. The dataset includes information about tournaments, dates, rounds, players, rankings, ATP points, playing conditions, betting odds and match scores.

Since the source data were stored in a single CSV file, they were reorganized into a reconciled relational database. This step separated repeated information about players, tournaments and rounds from the match records and made the relationships between the main entities explicit.

The fact selected for the multidimensional model is **MATCH**. Each fact instance represents one tennis match. The conceptual design followed a data-driven approach: the attribute tree was built from the reconciled schema, edited, and then used to identify the dimensions and measures of the Dimensional Fact Model (DFM).

The DFM was translated into a star schema centred on **FACT_MATCH**. The warehouse includes dimensions for date, tournament, round, surface, court, match format and player. The player dimension is used in four roles: winner, loser, favourite and underdog.

The project also includes data quality assessment, data cleaning and an ETL process from the reconciled database to the data warehouse. This report describes the data source, the reconciled schema, the cleaning decisions, the conceptual design, the logical star schema and the final ETL validation.

2 Data Source and Preliminary Requirement Analysis

2.1 Data source

The project is based on the *ATP Tennis Daily Pull* dataset available on Kaggle.¹

The file used for the project contains 67,512 men's professional tennis matches and 17 attributes. Each row represents one match.

The attributes describe:

- tournament name, date and tournament series;
- court type, surface, round and match format;
- the two players and the winner;
- rankings and ATP points of both players;
- pre-match betting odds;
- final score.

The CSV format is useful as a source, but it repeats player, tournament and round information in many rows and does not represent their relationships explicitly. For this reason, the data were first reorganized into a reconciled relational database.

¹<https://www.kaggle.com/datasets/dissfya/atp-tennis-2000-2023daily-pull/data>

2.2 Preliminary requirements

The main analytical objective is to study ATP matches, with particular attention to upsets. In this project, an upset is an observable match in which the winner has higher valid pre-match odds than the loser.

The preliminary requirements were:

- analyse the number of matches and upsets over time;
- compare the upset rate by surface, tournament series and round;
- study the relationship between results, rankings and ATP points;
- compare the odds of winners, losers, favourites and underdogs;
- support analysis at different levels of detail.

These requirements led to the selection of **MATCH** as the central fact. A match can be analysed by date, tournament, round, playing conditions and player role.

3 Reconciled Database

The original CSV file was reorganized into a reconciled relational database. The aim was to reduce redundancy, make the relationships between the main entities explicit and provide a consistent source for the data warehouse.

The reconciled database is composed of five tables: **PLAYER**, **TOURNAMENT**, **ROUND**, **MATCHES** and **MATCH_DERIVED**.

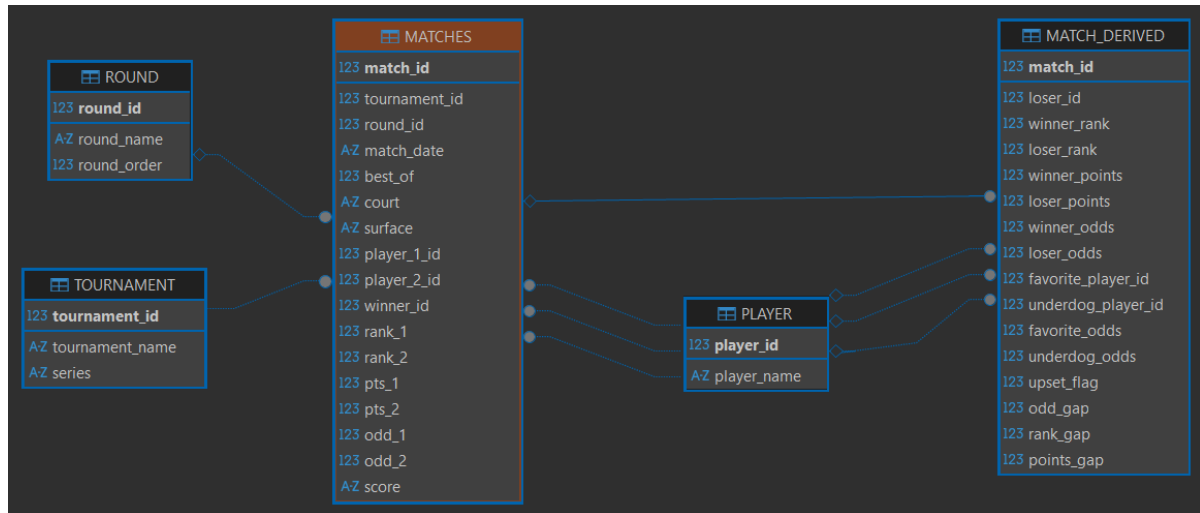


Figure 1: Schema of the reconciled database.

3.1 Main tables

Table	Description
PLAYER	Stores the unique identifier and name of each player.
TOURNAMENT	Stores the tournament identifier, tournament name and series.
ROUND	Stores the round identifier, round name and the ordering value used for the sequence of rounds.
MATCHES	Stores the source match information, including date, tournament, round, playing conditions, players, rankings, ATP points, betting odds and score.
MATCH_DERIVED	Stores attributes reorganized or derived for analysis, including winner and loser statistics, favourite and underdog roles, the upset indicator and the three gap measures.

Table 1: Main tables of the reconciled database.

3.2 Main relationships

Each record in **MATCHES** represents one tennis match. A match is linked to one tournament and one round through foreign keys.

The two participants and the winner are linked to **PLAYER**. The same table is therefore referenced with different roles.

The relationship between **MATCHES** and **MATCH_DERIVED** is one-to-one. For each match, the derived table stores attributes organised by winner, loser, favourite and underdog.

This separation is useful because the CSV mainly identifies the participants as Player 1 and Player 2, while the analysis is easier to express with role-based attributes such as winner rank, loser points, favourite odds and underdog odds.

The reconciled database is a relational representation of the source data. The star schema introduced later has a different purpose: it is designed for multidimensional analysis.

4 Data Quality Assessment and Cleaning

Data quality checks were carried out on the reconciled database before loading the data warehouse. The assessment covered completeness, uniqueness, consistency, numerical validity and referential integrity.

The source CSV uses -1 for several unavailable numerical values. In the reconciled database, these unavailable rankings, points and odds are stored as NULL. The single odds value equal to zero was kept so that it could be detected as an invalid value during the assessment.

4.1 Population and uniqueness checks

The number of records in each table is shown in Table 2.

Table	Number of rows
PLAYER	1,798
TOURNAMENT	331
ROUND	8
MATCHES	67,512
MATCH_DERIVED	67,512

Table 2: Population of the reconciled database.

Every record in `MATCHES` has one corresponding record in `MATCH_DERIVED`. No missing derived rows or orphan derived rows were found.

No duplicated `match_id` values were found in either table. Therefore, `match_id` identifies one match reliably.

4.2 Completeness checks

No missing values were found in the main attributes required to identify a match:

- `match_id`;
- `tournament_id`;
- `round_id`;
- `match_date`;
- `best_of`;
- `player_1_id`;
- `player_2_id`.

The missing values concern analytical attributes rather than the validity of the match record.

Check	Missing values
Winner ranking	0
Loser ranking	26
Ranking gap	26
Winner ATP points	15,652
Loser ATP points	15,653
Points gap	15,653
Winner odds	3,783
Loser odds	3,780
Matches with at least one missing odd	3,783

Table 3: Main completeness results.

All missing values of `rank_gap` are explained by a missing ranking. All missing values of `points_gap` are explained by missing ATP-point information.

4.3 Betting-odds checks

Betting odds require specific checks because they are used to identify the favourite, the underdog and the occurrence of an upset.

A total of 3,783 matches have at least one missing odd. A further 1,428 matches have equal winner and loser odds. These equal-odds cases consist of:

- 1,427 matches with positive equal odds;
- 1 match with both odds equal to 0.0.

The invalid case is match 65824, played on 22 July 2025. Since valid betting odds must be positive, the match was retained but excluded from betting-based analysis.

Overall, 5,211 matches do not allow a reliable distinction between favourite and underdog:

$$3783 + 1427 + 1 = 5211 \quad (1)$$

4.4 Consistency and validity checks

The loser, favourite and underdog identifiers were checked against the two players involved in each match.

No invalid records were found for:

- `loser_id`;
- `favorite_player_id`;
- `underdog_player_id`.

No case was found in which favourite and underdog were the same player.

In the reconciled database, no negative values were found for rankings or ATP points. The attribute `best_of` contains only 3 and 5, with 54,806 and 12,706 occurrences respectively.

4.5 Cleaning decisions

No valid match was deleted.

Missing rankings, ATP points and betting odds were kept as `NULL`. They were not replaced with averages, medians or constant values because imputation would create artificial values and affect:

- `rank_gap`;
- `points_gap`;
- `odd_gap`;
- `upset_rate`.

An upset is observable only when both winner and loser odds are available, positive and different:

$$\text{winner_odds} > 0 \quad \wedge \quad \text{loser_odds} > 0 \quad \wedge \quad \text{winner_odds} \neq \text{loser_odds} \quad (2)$$

The final counts are shown in Table 4.

Measure	Value
Total matches	67,512
Observable matches	62,301
Upsets	18,607
Non-upsets	43,694
Non-observable matches	5,211
Overall upset rate	29.87%

Table 4: Final upset-related results.

The upset rate is calculated only on observable matches:

$$\text{upset_rate} = \frac{\sum \text{upset_count}}{\sum \text{observable_upset_count}} \quad (3)$$

This prevents missing, equal or invalid odds from being counted as normal non-upsets.

5 Conceptual Design

The conceptual design was developed with a data-driven approach starting from the reconciled database. The main steps were the choice of the fact, the definition of its granularity, the construction and editing of the attribute tree, and the definition of the final DFM structure.

5.1 Choice of the fact

The selected fact is **MATCH**.

A match is the main event represented in the source. It occurs on a specific date, belongs to a tournament and a round, is played under defined conditions and involves two players. It is also associated with numerical properties such as rankings, ATP points and betting odds.

5.2 Fact granularity

The fact has transactional granularity:

One primary event corresponds to one tennis match.

Each row of the final fact table therefore represents exactly one source match.

5.3 Initial attribute tree

The initial attribute tree was built from **MATCHES**. It included the attributes stored directly in that table, the attributes reached through foreign keys and the attributes reached through the one-to-one relationship with **MATCH_DERIVED**.

The main branches are summarized in Table 5.

Branch	Attributes
Match information	<code>match_date</code> , <code>best_of</code> , <code>court</code> , <code>surface</code> , <code>score</code>
Tournament	<code>tournament_id</code> , <code>tournament_name</code> , <code>series</code>
Round	<code>round_id</code> , <code>round_name</code> , <code>round_order</code>
Players	<code>player_1_id</code> , <code>player_2_id</code> , <code>winner_id</code> , player names
Rankings and points	<code>rank_1</code> , <code>rank_2</code> , <code>pts_1</code> , <code>pts_2</code>
Betting odds	<code>odd_1</code> , <code>odd_2</code>
Derived attributes	loser, favourite and underdog identifiers, role-based rankings, ATP points, odds, gaps and <code>upset_flag</code>

Table 5: Main branches of the initial attribute tree.

At this stage, the tree was still close to the source schema and contained technical identifiers that were useful for navigation but not for OLAP aggregation.

5.4 Attribute-tree editing

The tree was edited through pruning, grafting and role reorganisation.

5.4.1 Pruning

Technical identifiers were removed from the final conceptual representation because they are join attributes rather than useful aggregation levels. These include:

- `match_id`;
- `tournament_id`;
- `round_id`;
- player identifiers;
- favourite and underdog identifiers.

The attribute `score` was removed because score analysis was not included in the final requirements.

The source-oriented pairs `rank_1/rank_2`, `pts_1/pts_2` and `odd_1/odd_2` were replaced by role-based attributes for winner, loser, favourite and underdog.

5.4.2 Grafting

After removing technical identifiers, their descriptive descendants were retained and connected to meaningful conceptual branches.

For example:

- `tournament_name` and `series` were retained as attributes of Tournament;
- `round_name` and `round_order` were retained as attributes of Round;
- player names were retained under the relevant player roles.

The technical one-to-one bridge between `MATCHES` and `MATCH_DERIVED` was removed from the final representation, while its analytical attributes were retained as properties of `MATCH`.

5.4.3 Player roles

Player 1 and Player 2 were not kept as final analytical roles. They were replaced by:

- Winner;
- Loser;
- Favourite;
- Underdog.

These four roles refer to the same player domain. Favourite and Underdog are optional because they cannot be identified in the 5,211 non-observable matches.

5.4.4 Date attributes

The attributes `month` and `year` were derived from `match_date`:

$$\text{match_date} \rightarrow \text{month} \quad \text{match_date} \rightarrow \text{year} \quad (4)$$

They support aggregation by calendar month and year. The implemented schema does not use `month` alone as a parent of `year`.

5.5 Final dimensions and attributes

The final dimensions are summarized in Table 6.

Dimension	Attributes
Date	<code>match_date</code> , <code>month</code> , <code>year</code>
Tournament	<code>tournament_name</code> , <code>series</code>
Round	<code>round_name</code> , with <code>round_order</code> used for ordering
Surface	<code>surface</code>
Court	<code>court</code>
Best of	<code>best_of</code>
Winner	<code>player_name</code>
Loser	<code>player_name</code>
Favourite	<code>player_name</code> , optional role
Underdog	<code>player_name</code> , optional role

Table 6: Dimensions and attributes of the `MATCH` fact.

Tournament name and series are kept as attributes of the Tournament dimension. No direct hierarchy `tournament_name` $\rightarrow$ `series` is assumed, because the same tournament name can occur with different series values in the source.

5.6 Measures

The additive counters are:

- `match_count`;
- `upset_count`;
- `non_upset_count`;
- `observable_upset_count`;
- `missing_upset_flag_count`;
- `equal_odds_count`.

They can be aggregated with `SUM` across the dimensions.

The remaining measures describe rankings, ATP points and betting odds:

- `winner_rank`, `loser_rank`, `rank_gap`;
- `winner_points`, `loser_points`, `points_gap`;
- `winner_odds`, `loser_odds`;
- `favorite_odds`, `underdog_odds`, `odd_gap`.

Their exact definitions are:

$$\text{rank_gap} = \text{loser_rank} - \text{winner_rank} \quad (5)$$

$$\text{points_gap} = \text{winner_points} - \text{loser_points} \quad (6)$$

$$\text{odd_gap} = \text{underdog_odds} - \text{favorite_odds} \quad (7)$$

These measures are not additive. Their meaningful aggregation operators are mainly `AVG`, `MIN` and `MAX`.

The derived measure `upset_rate` is calculated as:

$$\text{upset_rate} = \frac{\sum \text{upset_count}}{\sum \text{observable_upset_count}} \quad (8)$$

It is calculated at query time and is not stored in `FACT_MATCH`.

5.7 Final DFM fact schema

The final DFM is centred on `MATCH`. The fact is analysed by Date, Tournament, Round, Surface, Court, Best of and the four player roles.

DFM element	Content
Fact	MATCH
Granularity	One tennis match
Dimensions	Date, Tournament, Round, Surface, Court, Best of, Winner, Loser, Favourite, Underdog
Additive measures	Match and upset-related counters
Non-additive measures	Rankings, ATP points, odds and gap measures
Derived measure	upset_rate

Table 7: Summary of the final DFM fact schema.

The four player roles share the same player hierarchy. In the logical design, this is implemented with one role-playing dimension.

6 Logical Design

The DFM was translated into a relational star schema centred on **FACT_MATCH**. Descriptive attributes are stored in seven dimension tables.

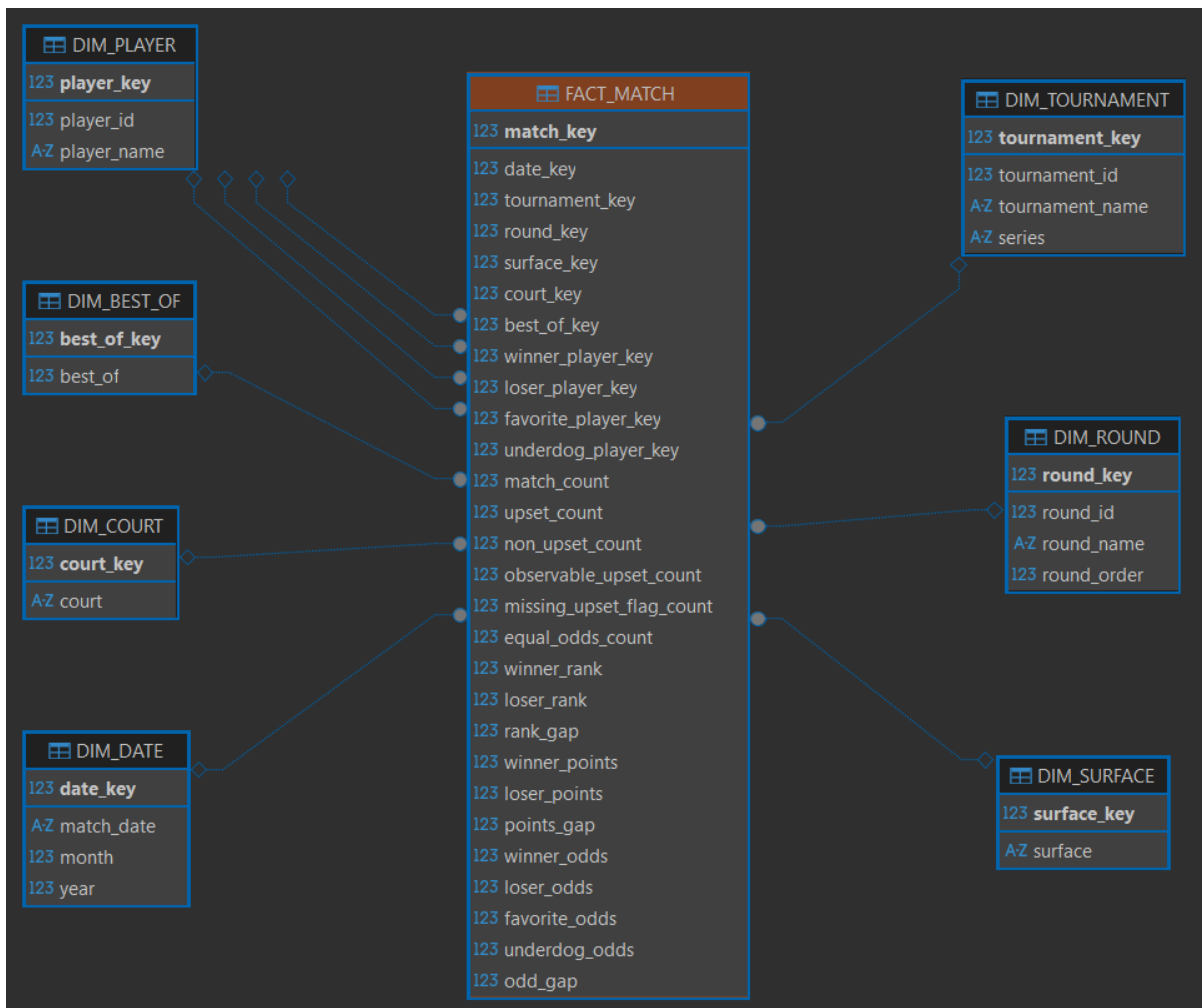


Figure 2: Logical star schema of the data warehouse.

6.1 Dimension tables

Dimension table	Main attributes
DIM_DATE	date_key, match_date, month, year
DIM_TOURNAMENT	tournament_key, tournament_id, tournament_name, series
DIM_ROUND	round_key, round_id, round_name, round_order
DIM_SURFACE	surface_key, surface
DIM_COURT	court_key, court
DIM_BEST_OF	best_of_key, best_of
DIM_PLAYER	player_key, player_id, player_name

Table 8: Dimension tables of the star schema.

DIM_TOURNAMENT, DIM_ROUND, DIM_SURFACE, DIM_COURT, DIM_BEST_OF and DIM_PLAYER use integer surrogate keys. In DIM_DATE, date_key is derived from the date in YYYYMMDD form.

The source identifiers are retained in DIM_TOURNAMENT, DIM_ROUND and DIM_PLAYER.

6.2 Fact table

FACT_MATCH keeps the transactional grain defined in the conceptual design. Each row represents one match.

The foreign keys are:

- date_key;
- tournament_key;
- round_key;
- surface_key;
- court_key;
- best_of_key;
- winner_player_key;
- loser_player_key;
- favorite_player_key;
- underdog_player_key.

During the ETL, match_key is populated with the source match_id. The favourite and underdog keys are NULL for the 5,211 non-observable matches.

6.3 Role-playing player dimension

DIM_PLAYER is referenced four times by FACT_MATCH, once for each role:

- winner;
- loser;
- favourite;
- underdog.

This is a role-playing dimension: one physical dimension table is used with different meanings according to the foreign key.

6.4 Glossary of measures

Table 9: Glossary of the measures in `FACT_MATCH`.

Measure	Definition	Aggregation	
<code>match_count</code>	Constant equal to 1 for each match.	SUM	
<code>upset_count</code>	Equal to 1 when valid unequal positive odds are available and the underdog is the winner.	SUM	
<code>non_upset_count</code>	Equal to 1 when valid unequal positive odds are available and the favourite is the winner.	SUM	
<code>observable_upset_count</code>	Equal to 1 when both winner and loser odds are available, positive and different.	SUM	
<code>missing_upset_flag_count</code>	Equal to 1 when favourite or underdog cannot be resolved because odds are missing, equal or invalid.	SUM	
<code>equal_odds_count</code>	Equal to 1 when both odds are positive and equal.	SUM	
<code>winner_rank</code>	Ranking position of the winner at match time.	AVG, MAX	MIN,
<code>loser_rank</code>	Ranking position of the loser at match time.	AVG, MAX	MIN,
<code>rank_gap</code>	Difference between loser ranking and winner ranking.	AVG, MAX	MIN,
<code>winner_points</code>	ATP points of the winner at match time.	AVG, MAX	MIN,
<code>loser_points</code>	ATP points of the loser at match time.	AVG, MAX	MIN,
<code>points_gap</code>	Difference between winner ATP points and loser ATP points.	AVG, MAX	MIN,
<code>winner_odds</code>	Pre-match odds associated with the winner.	AVG, MAX	MIN,
<code>loser_odds</code>	Pre-match odds associated with the loser.	AVG, MAX	MIN,
<code>favorite_odds</code>	Lower valid pre-match odds.	AVG, MAX	MIN,
<code>underdog_odds</code>	Higher valid pre-match odds.	AVG, MAX	MIN,
<code>odd_gap</code>	Difference between underdog odds and favourite odds.	AVG, MAX	MIN,
<code>upset_rate</code>	Ratio between total upsets and total observable matches. Not physically stored.	Derived ratio	

7 ETL Process and Final Validation

The ETL process was implemented in Google Colab. Python and SQL were used to extract data from the reconciled database, transform them and load the dimensional tables of the data warehouse.

Before loading, the existing rows in the fact and dimension tables were deleted so that the notebook could be executed again without retaining rows from a previous load.

7.1 Loading the dimensions

The dimensions were loaded before the fact table. The executed notebook produced the row counts shown in Table 10.

Table	Rows
DIM_DATE	6,609
DIM_TOURNAMENT	331
DIM_ROUND	8
DIM_SURFACE	4
DIM_COURT	2
DIM_BEST_OF	2
DIM_PLAYER	1,798

Table 10: Rows loaded into the dimension tables.

For DIM_DATE, `date_key` was generated as the integer representation of the date in YYYYMMDD form. Month and year were extracted from `match_date`.

The small categorical dimensions were populated from the distinct values of surface, court and best-of format.

7.2 Loading the fact table

After the dimensions, FACT_MATCH was loaded by resolving source identifiers and categorical values into dimension keys.

DIM_PLAYER was joined four times to resolve winner, loser, favourite and underdog. `match_key` was set equal to the source `match_id`.

The fact table contains 67,512 rows, exactly the same number as MATCHES.

Check	Result
Reconciled matches	67,512
Fact-table rows	67,512
Difference	0

Table 11: Comparison between reconciled matches and fact rows.

7.3 Validation of measures

The final measure values are:

- 67,512 total matches;
- 62,301 observable matches;
- 18,607 upsets;
- 43,694 non-upsets;
- 5,211 non-observable matches;

- an upset rate of 0.2987, corresponding to 29.87%.

The following consistency condition was verified:

$$\sum \text{observable_upset_count} = \sum \text{upset_count} + \sum \text{non_upset_count} \quad (9)$$

Both sides are equal to 62,301.

The total number of matches was also verified as:

$$62301 + 5211 = 67512 \quad (10)$$

7.4 Foreign-key validation

No missing values were found in the mandatory keys for date, tournament, round, surface, court, best-of format, winner and loser.

The favourite and underdog keys each contain 5,211 missing values. These correspond to the non-observable betting cases.

The final notebook health check returned OK for:

- fact-table row count;
- observable-match consistency;
- non-observable-match consistency;
- mandatory foreign keys.

7.5 Export for Tableau

The fact table and all seven dimension tables were exported as CSV files. A flat export was also created by joining the fact table with its dimensions.

The file `tableau_match_flat.csv` contains 67,512 rows and 34 columns. It includes match descriptors, player names and measures.

The missing values in the flat export match the data-quality findings: 26 missing loser rankings and rank gaps, 15,652 missing winner points, 15,653 missing loser points and points gaps, 3,783 missing winner odds, 3,780 missing loser odds, and 5,211 missing favourite/underdog names and odds-related derived values.

8 Conclusions

The project transformed a flat tennis dataset into a data warehouse suitable for multidimensional analysis.

The source data were first reorganized into a reconciled relational database. The fact **MATCH** was selected with a transactional grain of one match. The conceptual design identified the dimensions, player roles and measures, and the logical design produced a star schema with one fact table and seven dimension tables.

The data quality assessment found missing rankings, ATP points and betting odds, together with one invalid zero-odds case. No valid match was removed. Missing values were preserved when no reliable value was available, and upset-related measures were calculated only for observable matches.

The executed ETL loaded all 67,512 matches into `FACT_MATCH`. The final checks confirmed the row counts, measure consistency and mandatory foreign keys. The resulting warehouse supports analysis by time, tournament, round, surface, court, match format and player role.

The resulting warehouse supports OLAP operations such as roll-up, drill-down, slice and dice across the available dimensions.